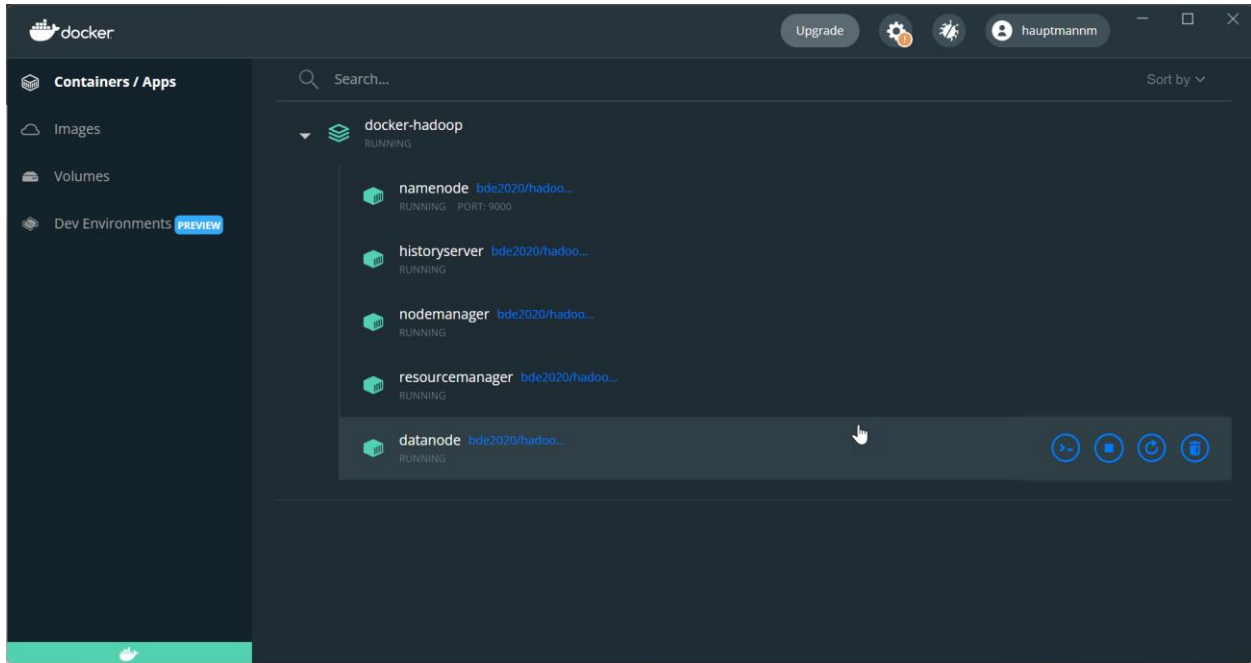


Activity 18.2 Solution File

1.



2.

Name	Type	Compressed size	Password pr...	Size	Ratio	Date modified
2701-0	Text Document	501 KB	No	1,247 KB	60%	8/18/2021 7:41 AM

3.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
# ls -l
total 1104
-rw-r--r-- 1 root root 325083 Feb  4 2020 KEYS
drwxr-xr-x 1 root root 4096 Feb  4 2020 bin
drwxr-xr-x 2 root root 4096 Sep  8 2019 boot
drwxr-xr-x 5 root root 340 Dec 17 11:06 dev
-rwxr-xr-x 1 root root 4155 Feb  4 2020 entrypoint.sh
drwxr-xr-x 1 root root 4096 Dec 17 11:06 etc
drwxr-xr-x 3 root root 4096 Feb  4 2020 hadoop
drwxr-xr-x 2 root root 4096 Feb  4 2020 hadoop-data
-rw-r--r-- 1 root root 697323 Dec 18 15:39 hadoop-mapreduce-examples-2.7.1-sources.jar
drwxr-xr-x 2 root root 4096 Sep  8 2019 home
drwxr-xr-x 2 root root 4096 Dec 18 17:24 input
drwxr-xr-x 1 root root 4096 Jan 30 2020 lib
drwxr-xr-x 2 root root 4096 Jan 30 2020 lib64
drwxr-xr-x 2 root root 4096 Jan 30 2020 media
drwxr-xr-x 2 root root 4096 Jan 30 2020 mnt
drwxr-xr-x 1 root root 4096 Feb  4 2020 opt
drwxr-xr-x 2 root root 4096 Dec 17 13:03 output
dr-xr-xr-x 246 root root  0 Dec 17 11:06 proc
drwx----- 1 root root 4096 Dec 17 11:43 root
drwxr-xr-x 3 root root 4096 Jan 30 2020 run
-rwxr-xr-x 1 root root 494 Feb  4 2020 run.sh
drwxr-xr-x 1 root root 4096 Feb  4 2020 sbin
drwxr-xr-x 2 root root 4096 Jan 30 2020 srv
dr-xr-xr-x 11 root root  0 Dec 17 11:06 sys
drwxrwxrwt 1 root root 4096 Dec 18 17:25 tmp
drwxr-xr-x 1 root root 4096 Jan 30 2020 usr
drwxr-xr-x 1 root root 4096 Jan 30 2020 var
#
```

4.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
# ls -l input
total 1248
-rwxr-xr-x 1 root root 1276235 Dec 17 12:43 2701-0.txt
#
```

5.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
# hadoop fs -mkdir -p input
#
```

6.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
# hdfs dfs -put ./input/* input
2021-12-20 12:03:08,758 INFO sas1.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteHostTrusted = false
#
```

7.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
# curl -L https://repo1.maven.org/maven2/org/apache/hadoop/hadoop-mapreduce-examples/2.7.1/hadoop-mapreduce-examples-2.7
.1-sources.jar --output hadoop-mapreduce-examples-2.7.1-sources.jar
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 680k  100 680k    0     0  1167k      0 --:--:-- --:--:-- --:--:-- 1168k
#
```

8.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
Total megabyte-milliseconds taken by all map tasks=8880128
Total megabyte-milliseconds taken by all reduce tasks=19619840
Map-Reduce Framework
  Map input records=22316
  Map output records=215864
  Map output bytes=2113253
  Map output materialized bytes=159938
  Input split bytes=112
  Combine input records=215864
  Combine output records=33568
  Reduce input groups=33568
  Reduce shuffle bytes=159938
  Reduce input records=33568
  Reduce output records=33568
  Spilled Records=67136
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=61
  CPU time spent (ms)=2750
  Physical memory (bytes) snapshot=600764416
  Virtual memory (bytes) snapshot=13559898112
  Total committed heap usage (bytes)=473432064
  Peak Map Physical memory (bytes)=359907328
  Peak Map Virtual memory (bytes)=5107712000
  Peak Reduce Physical memory (bytes)=240857088
  Peak Reduce Virtual memory (bytes)=8452186112
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=1276235
File Output Format Counters
  Bytes Written=372792
#
```

9.

```
docker exec -it 79162100cf0182e9daceabac66e19652ddbde959d94c76c4fa3feeb8bd6ae143 /bin/sh
""My 1
""Nay, 2
""Nay," 1
""Oh 1
""Say 1
""Shall 1
""Shut 1
""Sink 1
""So 2
""Stern 1
""The 1
""Then 2
""This 1
""Though 1
""Turn 3
""Very 1
""Well 1
""What 3
""Where 1
""Who's 1
""Why 1
""Will 2
""Yes, 1
""You 2
""Bout 1
""Hind 1
""Tis 2
""Twill 1
""tis 1
root@79162100cf01:/# 6~5~
```